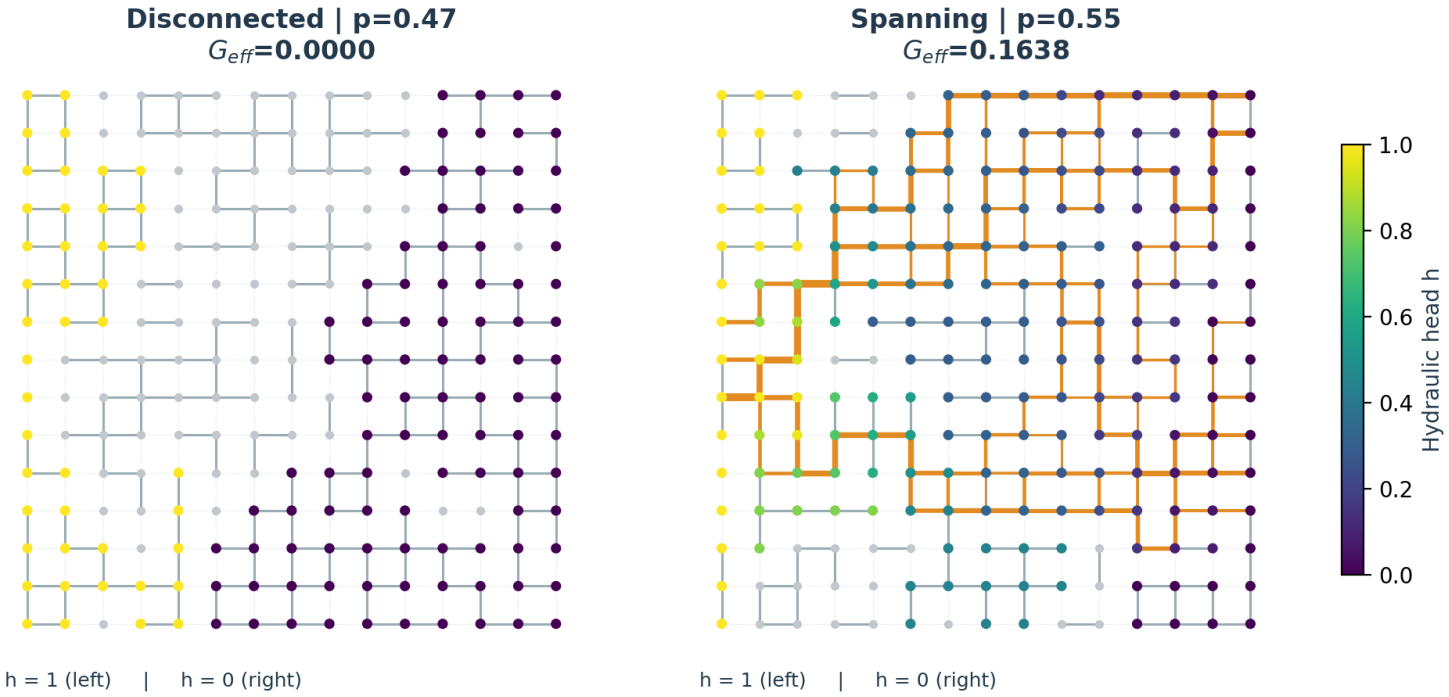
Flow in a 2D random network

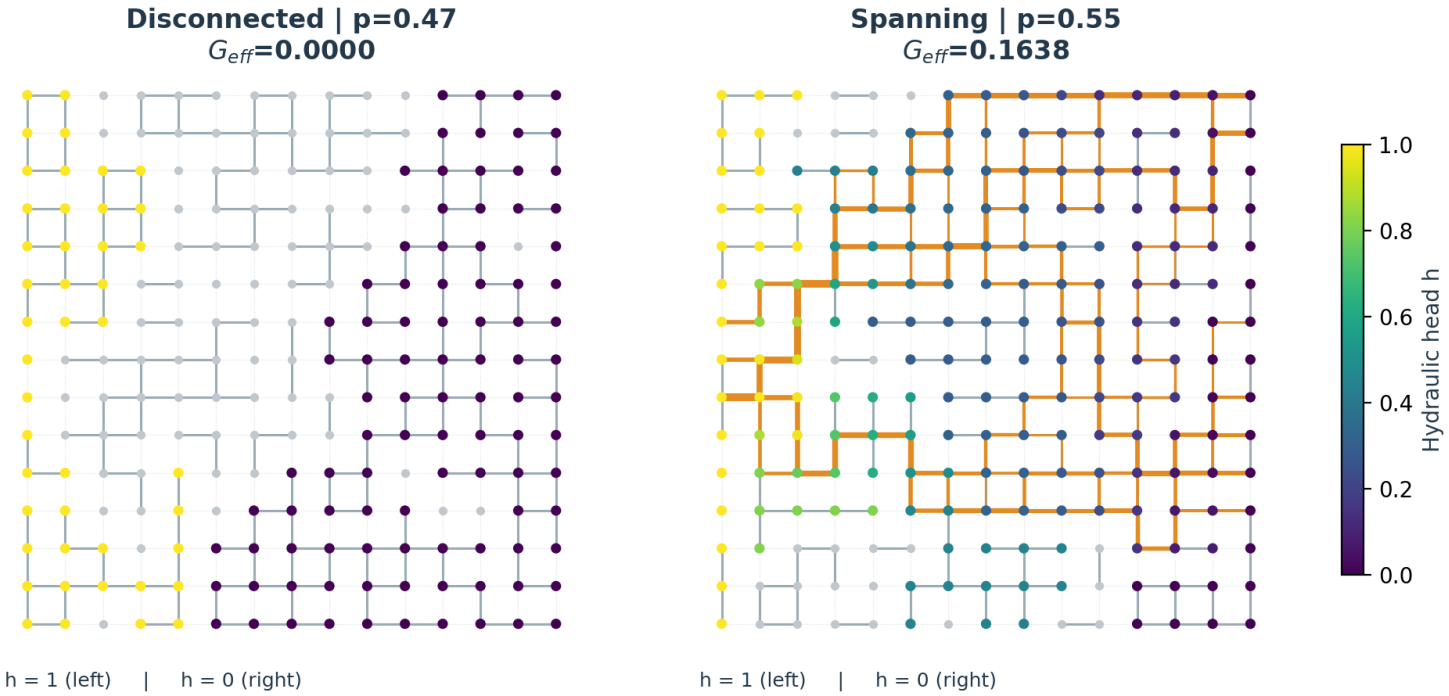
Uncertainty, global sensitivity and verified computation



- Connectivity dominates flow variability over the selected ranges.
- 248,188 model evaluations; 24 independent verification tests.
- Completed analysis • 04 November 2026

A hierarchical random-network model

625 nodes · 1,200 possible links · n = 25



- Draw ONE p per realization: p = 0.5 + δU, U uniform on [−1, 1].
- Conditional on p: each link active independently with probability p.
- Active links: ln(g) ~ Normal(0, σ²); inactive links: g = 0.
- Left head = 1; right head = 0; no flow across top and bottom.

Solve only hydraulically relevant components

Sparse Kirchhoff equations with explicit disconnected-component treatment

Verification	Evidence
Uniform grid	G = g n/(n−1), exact linear head
Series / parallel branches	Analytical resistance identities
Conservation / energy	Qin = Qout = dissipated energy
Scaling / monotonicity	G(cg)=cG(g); Rayleigh property
Ishigami benchmark	Known first and total Sobol indices
1,000 random diagnostics	Max residual 2.75e-13

- Find components that touch both fixed-head boundaries.
- Solve $L_{FF} h_F = -L_{FB} h_B$ on their free nodes.
- One-boundary clusters: constant head. Floating islands: arbitrary head, zero flow.
- G_{eff} = inlet flow / Δh; here $\Delta h = 1$.

An explicit, reproducible experiment

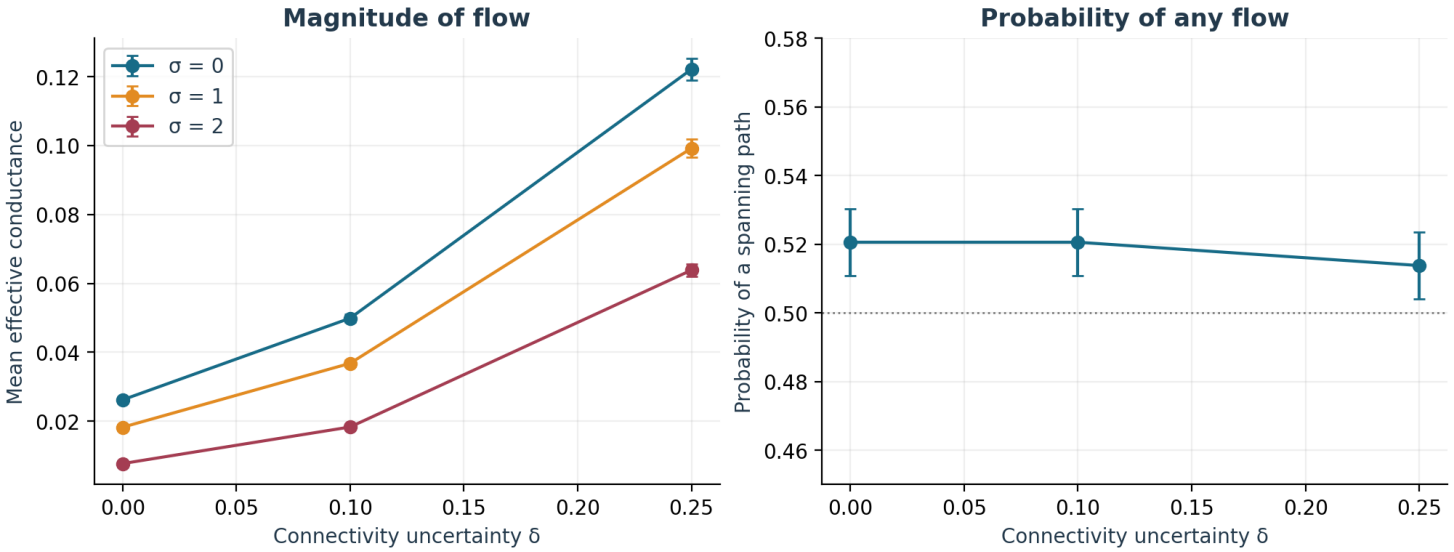
Separate RNG streams; common random numbers for paired scenario comparisons

Experiment	Design	Evaluations
Main uncertainty	9 × 10,000	90,000
Fixed-p sweep	9 × 2,500	22,500
Extra sizes	4 × 5,000	20,000
Grouped Sobol	16,384 × 7	114,688
Numerical diagnostics	1,000	1,000
Total	Size 25 reuses main data	248,188

- $\delta = 0, 0.10, 0.25$; $\sigma = 0, 1, 2$.
- Compare n = 15, 25 and 41 to expose finite-size dependence.
- Seed 20261104; all realization-level outputs are retained.

More uncertain connectivity raises mean flow

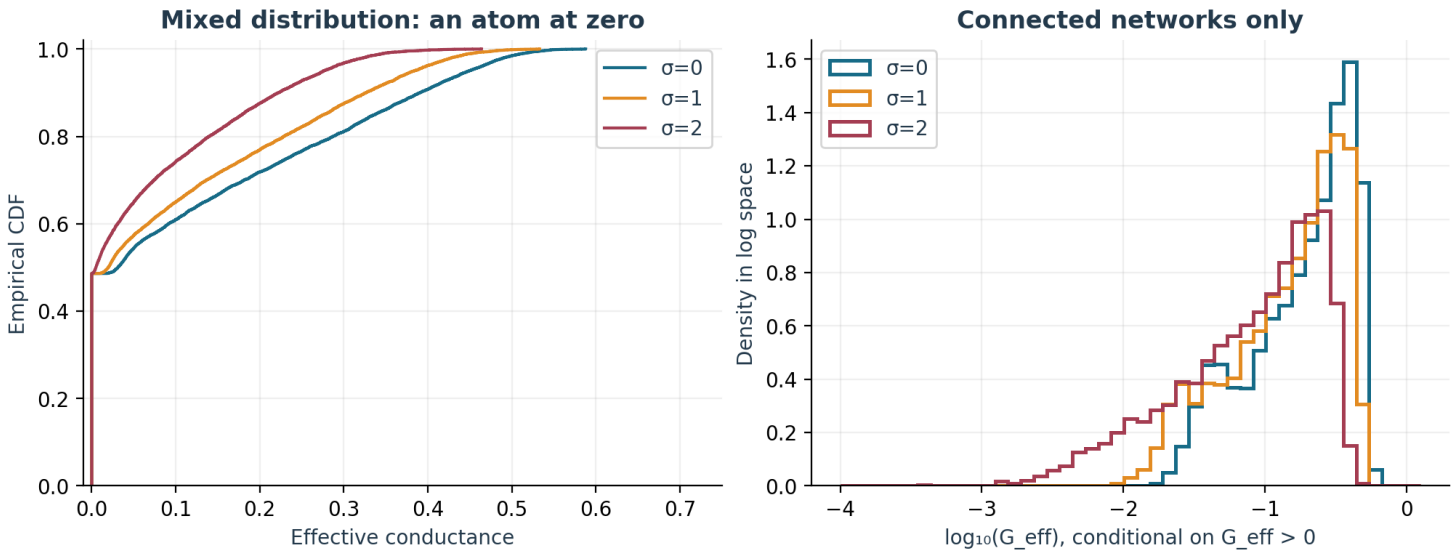
The probability of any flow remains close to one half



- At σ=1, mean G: 0.0183 → 0.0992 as δ: 0 → 0.25.
- Mean ratio: 5.44×, despite unchanged E[p]=0.5.
- Reference spanning probability: 51.38%.
- High-p realizations contribute substantial flow; low-p realizations often remain at zero.

A mean cannot describe the output distribution

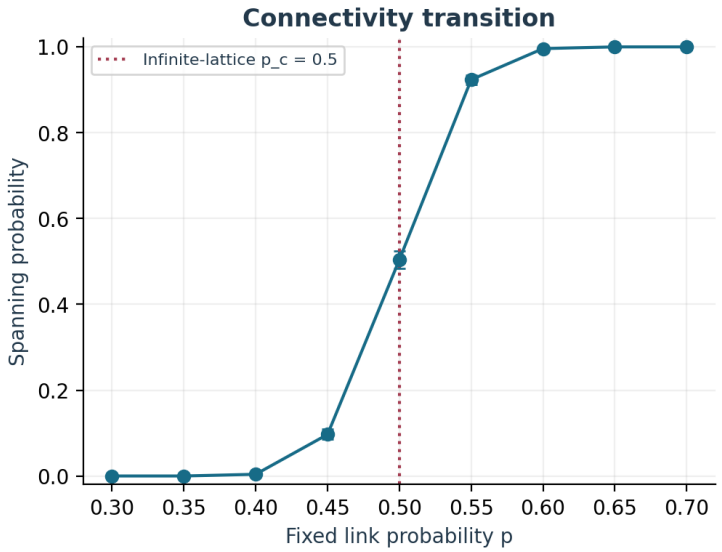
Separate zero-flow probability from positive-flow statistics

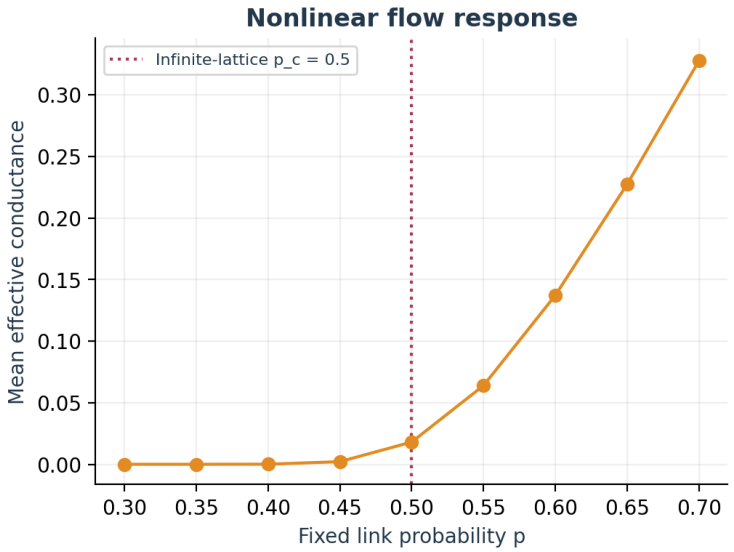


- Reference: δ=0.25, σ=1; P(G=0) = 48.62%.
- Mean 0.0992; median 0.0195; 95th percentile 0.3838.
- Mean given G>0: 0.1931.
- A single lognormal law is unsuitable for a distribution with an atom at zero.

A transition controls the response

Fixed-p sweep: 2,500 realizations per p; σ=1; n=25





- For an infinite square bond lattice, p_c = 0.5 (Kesten, 1980).
- A finite network has a smooth connectivity crossover.
- Nonlinear flow response explains the strong effect of the width δ.
- Shared p induces covariance $\delta^2/3$ between distinct link indicators.

Sensitivity of a stochastic model

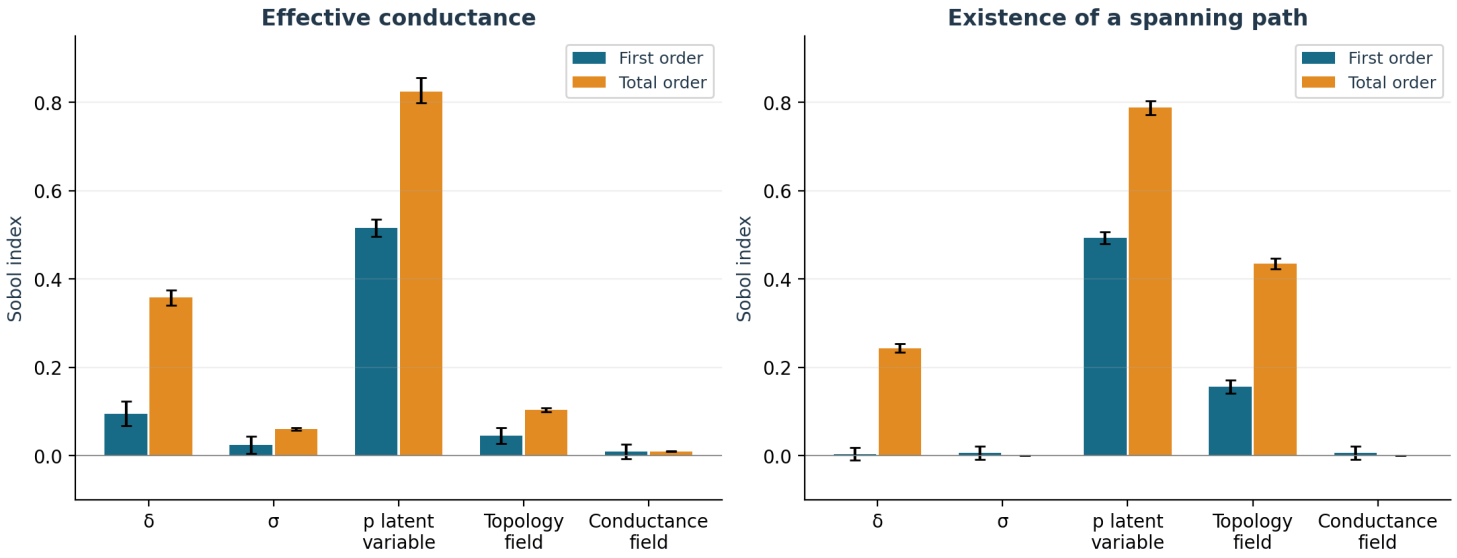
Grouped Sobol analysis on independent latent inputs

Group	Distribution / role
δ	Uniform(0, 0.25), connectivity width
σ	Uniform(0, 2), log-conductance SD
U	Uniform(−1, 1), p = 0.5 + δU
Topology W	1,200 independent Uniform(0,1) values
Conductance Z	1,200 independent standard normals

- Use five independent groups; keep δ and realized p out of the same independent-input list.
- Jansen first and total indices: interactions are explicitly retained.
- 16,384 base pairs; 1,000 paired-row bootstrap resamples.
- The estimator also passes the analytic Ishigami benchmark.

Global connectivity is the dominant influence

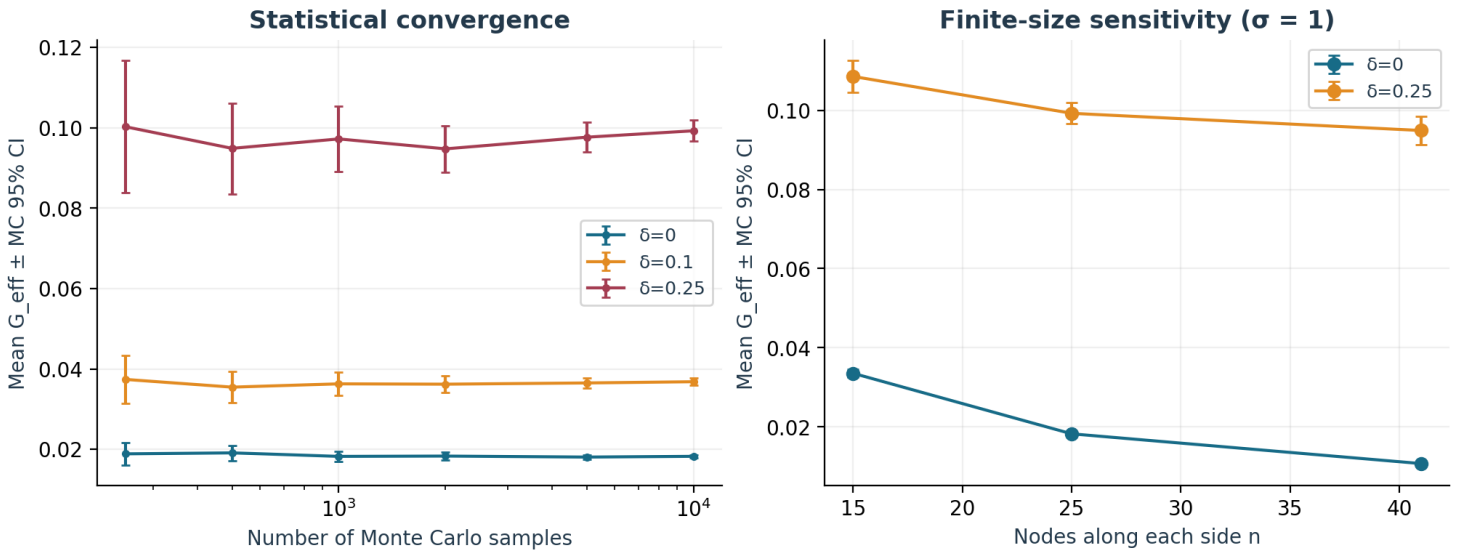
Total-order indices include every interaction involving each group



- U: total 0.825; first 0.516.
- δ: total 0.358; first 0.096. Strong interactions.
- Topology: 0.104; σ: 0.061; Z: 0.010.
- σ and Z have exactly zero total effect on path existence. Do not sum overlapping total indices.

Sampling precision improves; size effects remain

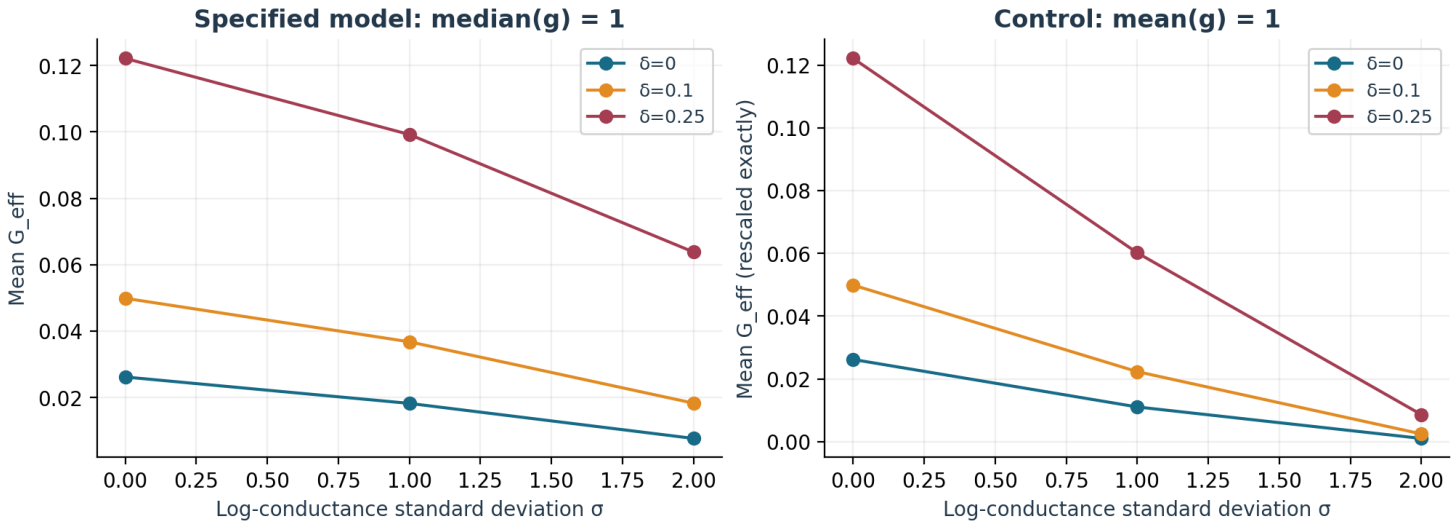
Monte Carlo convergence is distinct from finite-size sensitivity



- Final relative 95% half-widths: 2.19–2.91% for scenario means.
- At δ=0, σ=1: mean G decreases substantially with n.
- Changing n changes the random-network ensemble.
- A size-independent continuum limit has not been demonstrated.

Heterogeneity is not a pure variance change

The specified lognormal family fixes the median, not the mean



- Active-link $E[g] = \exp(\sigma^2/2)$: 1, 1.6487, 7.3891 for $\sigma = 0, 1, 2$.
- Mean network conductance still falls with σ over these scenarios.
- Weak links on series paths act as bottlenecks.
- Constant-mean control: $g^* = g \exp(-\sigma^2/2)$, so $G^* = G \exp(-\sigma^2/2)$ exactly.

Interpretation and next steps

Numerically verified behavior of a simplified dimensionless lattice

Deliverable	Location
Persian and English reports	reports/report_fa.pdf; report_en.pdf
Code and 24 tests	src/flow_network; tests/
Raw samples and statistics	results/
Executed notebook	notebooks/analysis.ipynb
One-command reproduction	README.md; scripts/
GitHub	mohrezsayeh90-ai/Interview_upsla

- Measure connectivity carefully; report connection probability and flow magnitude separately.
- Limits: regular 2D geometry, exploratory priors, steady linear single-phase flow, finite size.
- Next: 3D fracture geometry, aperture-based conductance, correlation, matrix exchange and calibration.
- No field-data validation or geological prediction accuracy is claimed.